Thermodynamic State Vector Interfaces and Exergy Destruction Mechanics in the Web of Life Architecture

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Sprint 003 Technical Preprint

Abstract

We present the formal specification and software implementation of Sprint 003 for the *Web of Life* computational ecosystem: the Thermodynamic State Vector Interface (`src/thermodynamics/types.ts` Grounded in non-equilibrium thermodynamics and systems ecology, this architectural milestone establishes rigorous mathematical contracts for mass and energy conservation (First Law) alongside explicit tracking of internal entropy generation rates ($\dot{S}_{\text{gen}} \geq 0$) and exergy destruction rates ($\dot{I} = T_0 \dot{S}_{\text{gen}}$, Second Law). By codifying boundary flux arrays and invariant validation predicates into pure TypeScript monad transitions, the framework ensures thermodynamic admissibility across Earth system compartments (*EarthPods*). This preprint outlines the governing equations, interface definitions, and verification strategies that bridge macroscopic thermodynamic theory with executable ecological software.

Official Repository: `https://github.com/pascalranoroarijaona/WebOfLife`

1 Introduction and Thermodynamic Foundations

Complex ecological and planetary systems operate as open, far-from-equilibrium thermodynamic structures driven by external solar irradiance and bounded by longwave radiative cooling to space. To model these dynamics without violating fundamental physical laws, simulation software must enforce strict accounting of energy conservation, mass fluxes, and irreversible entropy production.

Sprint 003 formalizes these requirements within the *Web of Life* codebase by introducing strict contractual types, state vector interfaces, and invariant checks in `src/thermodynamics/types.ts`. Building upon our foundational thermodynamic structures (`src/thermodynamics/thermodynamic_structure.` this sprint ensures that every state transition executed by simulation runloops adheres to:

1. **The First Law of Thermodynamics:** Conservation of total mass and internal energy within any control volume.
2. **The Second Law of Thermodynamics:** Non-negative internal entropy generation ($\dot{S}_{\text{gen}} \geq 0$) and proportional exergy destruction ($\dot{I} = T_0 \dot{S}_{\text{gen}}$).
3. **Solar-Driven Boundary Conditions:** Strict solar energy inputs coupled with terrestrial and atmospheric radiative loss pathways.

2 Mathematical Formulation of Thermodynamic Monads

2.1 First Law: Mass and Energy Conservation

For any control volume CV, the net mass change over time step Δt is governed by species mass flux divergences across the boundary:

$$\frac{dM_{\text{total}}}{dt} = \sum_i \dot{m}_{\text{in},i} - \sum_j \dot{m}_{\text{out},j} \tag{1}$$

Internal energy U evolution accounts for net radiative fluxes, sensible and latent convective exchanges, and mass-carried enthalpy:

$$\frac{dU}{dt} = \dot{Q}_{\text{radiative}} + \dot{Q}_{\text{convective}} + \sum \dot{m}_{\text{in}} h_{\text{in}} - \sum \dot{m}_{\text{out}} h_{\text{out}} \tag{2}$$

2.2 Second Law: Entropy Generation & Exergy Destruction

Irreversibilities within control volumes generate entropy at a rate $\dot{S}_{\text{gen}} \geq 0$. From the Clausius inequality:

$$\dot{S}_{\text{gen}} = \frac{dS}{dt} - \left(\sum_k \frac{\dot{Q}_k}{T_k} + \sum_{\text{in}} \dot{m}_{\text{in}} s_{\text{in}} - \sum_{\text{out}} \dot{m}_{\text{out}} s_{\text{out}} \right) \geq 0 \tag{3}$$

The corresponding **Exergy Destruction Rate** ($\dot{I}$) relative to ambient reference temperature T_0 is defined as:

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \tag{4}$$

3 Software Architecture & Interface Specifications

The implementation in `src/thermodynamics/types.ts` defines pure TypeScript interfaces enforcing these mathematical invariants:

```
export interface BoundaryFluxArray {
  readonly radiativeFlux: number;       // [W]
  readonly convectiveFlux: number;      // [W]
  readonly massEnthalpyFlux: number;    // [W]
  readonly speciesMassFluxes: Record<string, number>; // [kg/s]
}

export interface EntropyMetrics {
  readonly sGenRate: number;              // [W/K], must be >= 0
  readonly referenceTemperature: number;  // [K]
  readonly exergyDestructionRate: number; // [W], must match T_0 * sGenRate
}

export interface ThermodynamicStateVector {
  readonly timestamp: number;
  readonly internalEnergy: number;      // [J]
  readonly totalMass: number;           // [kg]
  readonly temperature: number;         // [K]
  readonly fluxes: BoundaryFluxArray;
  readonly entropyMetrics: EntropyMetrics;
}
```

State transitions operate as pure functions (`ThermodynamicTransitionFunction`) subject to strict invariant validation predicates (`validateThermodynamicInvariants`), which immediately reject unphysical states violating the Second Law or conservation principles.

4 Verification, Testing, and Conclusion

Sprint 003 establishes a robust testing paradigm (`tests/sprint_003.test.ts`) verifying:

- Immediate rejection of state vectors where $\dot{S}_{\mathrm{gen}} < 0$ or $\dot{I} < 0$.
- Mathematical consistency between exergy destruction and ambient-scaled entropy generation ($|\dot{I} - T_0 \dot{S}_{\mathrm{gen}}| < 10^{-5}$).
- Preservation of mass and energy across open system EarthPods.

By embedding rigorous thermodynamic constraints directly into the type system, *Web of Life* ensures that emergent ecological phenomena remain strictly tethered to physical reality.

References

[1] Bejan, A. (2006). *Advanced Engineering Thermodynamics*. John Wiley & Sons.

[2] Swenson, R. (1989). Emergent evolutionary systems: A thermodynamic explanation of innovation and structure. *Systems Research*, 6(3), 187–197.